

DartGrid III: A Semantic Grid Toolkit for Data Integration

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Abstract

A toolkit called DartGrid¹ using technologies from either semantic web or grid is introduced. It is build upon Globus Toolkit and uses RDF-based ontology to integrate heterogeneous relational databases. Technically, database resources are exposed as grid services and organized as an ontology-based virtual organization. By uniformly defined domain semantics, data resources are semantically registered to an semantic registration service and seamlessly integrated together to provide unified semantic query and semantic search service. Furthermore, a set of semantic tools are developed to raise the level of interaction with the system to a semantic level. A visual mapping tool is offered to help data provider to define semantic mapping from relational schema to shared ontology. A visual query tool which can automatically generated query interface based on ontology definitions, is provided to help specify complex semantic queries. A search-engine-like interface is offered to enable quickly search and semantically navigate data from one database to another database. DartGrid is a deployed system that currently provides query and search service at China Academy of Traditional Chinese Medicine².

1. Introduction

The *Semantic Web* is a exciting evolution of current web of documents, linked by hyper-link, into a web of data objects, linked by semantic link, providing more accurate query and search like one huge global database [1]. It is based upon *Resource Description Framework (RDF)*, a language for representing information in a minimally constrained, flexible, meaningful way, allowing data to be exchanged and integrated without loss of semantics, to be reused and shared across application, enterprize, and com-

munity boundaries.

The *Grid* community are working on the next generation of network architecture and distributed computing infrastructure enabling coordinated resource sharing and problem solving in dynamic, multi-institutional virtual organizations [2].

The *Semantic Grid* integrates the work on grid architecture from grid community and the work on web semantics from semantic web area, aiming to provide an interconnection environment that can effectively organize, share, cluster, fuse, and manage globally distributed versatile resources based on the interconnection semantics [3].

DartGrid is a semantic grid toolkit for data integration using technologies from either semantic web and grid. In a nutshell, it has the following technical features:

- Database resources are exposed as grid services and organized as an ontology-based virtual organization.
- By uniformly defined domain semantics, data resources are semantically registered to an semantic registration service and seamlessly integrated together to provide unified semantic query and semantic search service.
- A visual mapping tool is offered to help data provider to define semantic mapping from relational schema to shared ontology.
- A visual query tool which can automatically generated query interface based on ontology definitions, is provided to help specify complex semantic queries.
- A search-engine-like interface is offered to enable quickly search and semantically navigate data from one database to another database.

DartGrid can be used to integrate data in a specific *Virtual Organization*. The data sources may be of different logical structures or physical properties, and distributively locate at different places. But the user of DartGrid can

¹DartGrid Official Website: <http://ccnt.zju.edu.cn/projects/dartgrid>

²TCMGrid Official Website: <http://coworklab.cintcm.com/tcmsearch/>

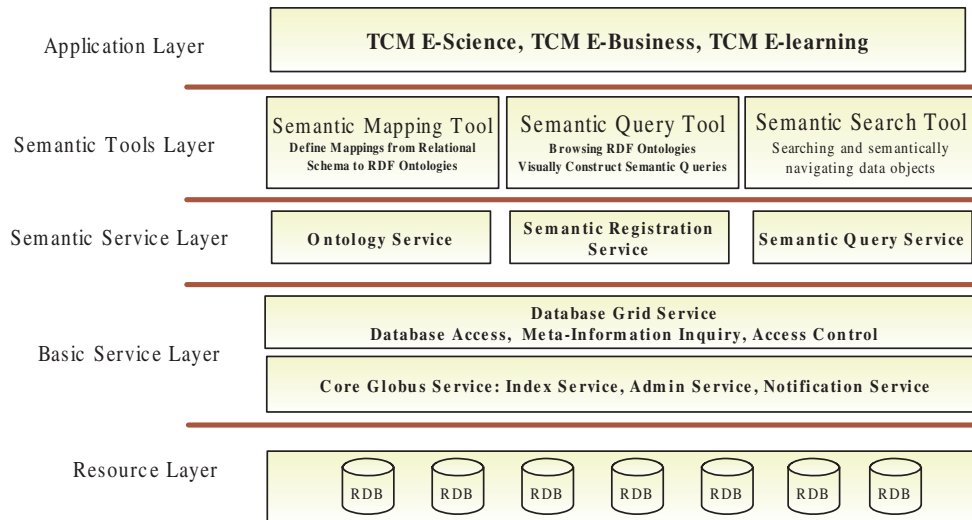


Figure 1. Layered Architecture of DartGrid.

query and search data without knowledge of any specific data source. All they need to be aware of is the ontology schema (or standard glossary) defined by domain experts or VO administrator.

Typically, a domain ontology is defined for a specific virtual organization. Database providers use a semantic registration tool to define the mappings from relational schema to domain ontology, and submit the registration entry to a semantic registration service. Normal user browses the ontology and specifies a semantic query by a query interface generated automatically according to the domain ontology. Programmer can encode a semantic query (a RDF query) into his programs and query data by some semantic interface exposed as grid service. When a semantic query is submitted, it will be translated into a set of SQL queries based on the semantic mapping information from relational schema to shared ontology. Next, data is retrieved from distributed databases and merged as one unified format to return.

Finally, DartGrid is a truly deployed system that currently provides query service at China Academy of Traditional Chinese Medicine.

This paper is outlined as below: Section 2 presents the general architecture of DartGrid. Section 3

2 General Architecture

2.1 Layered Service Architecture

Figure 1 illustrates the layered architecture of DartGrid. At the basic service layer, three services are implemented.

1. **Database Access Service.** It supports the typical remote operations on database contents, including query, insertion, deletion, and modification.
2. **Database Information Service.** It supports inquiring about the meta information of databases. The meta information includes: relational schema definition, DBMS descriptions, privilege information, statistics information including CPU utilization, available storage space, active session number etc..
3. **Access Control Service.** This service provides the service of authorizing or authenticating a grid user to access specific database resource.

We mainly contribute at the semantic service level. The services at this level are designed for RDF-based relational schema mediation and semantic query processing.

1. **Ontology Service.** This service is used to expose the shared ontologies that are defined using RDF/OWL languages. The ontologies are used to mediate heterogeneous relational databases.
2. **Semantic Registration Service.** Semantic registration establishes the mappings from source relational schema to mediated RDF ontologies. Semantic Registration Service maintains the mapping information and provides the service of registering and inquiring about this information.
3. **Semantic Query Service.** This service accepts RDF semantic queries. When a semantic query is submitted, several things DartGrid will do are:

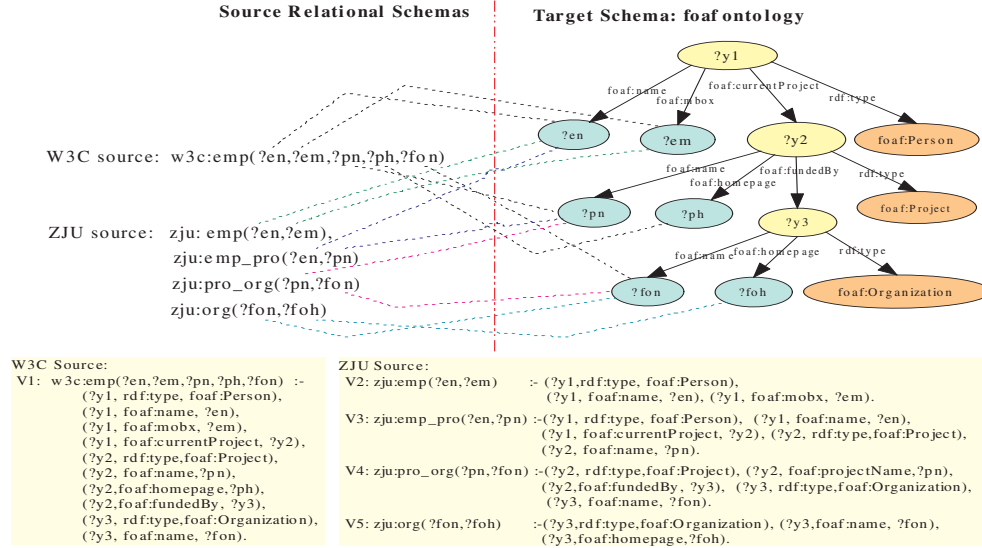


Figure 2. RDF View Example. The symbols `?en`, `?em`, `?pn`, `?ph`, `?fon`, `?foh` are variables and represent, respectively, "employee name", "mail box", "project name", "project homepage", "funding organization name", "funding organization homepage". The foaf ontology consists of three classes: `foaf:Person`, `foaf:Project`, `foaf:Organization`

- Parse the semantic query request.
- Translate the semantic query into a SQL query plan using the information from semantic registration service.
- Get the data resources' information and dispatch the query plan to different data resources.
- Collect the results returned, form a query result as RDF form.
- (Optional) Wrap the result into a user defined way.

3 Semantic Mapping

DartGrid uses RDF-based ontology to mediate heterogeneous relational database schemas. Fundamentally, DartGrid takes a view-based approach [4] to define the semantic mapping from source relational schema to mediated RDF ontologies, and offer a visual mapping tool to help define mappings.

3.1 Semantic Mapping and RDF Views

Consider a simple example: suppose both W3C and Zhejiang University (abbreviated as ZJU) have a legacy relational database about their employees and projects, and we

would like to integrate them by the FOAF ontology³, so that we can query these relational databases by formulating RDF queries upon the FOAF ontology.

The mapping scenario in Fig. 2 illustrates two source relational schemas (W3C, and ZJU), a target RDF schema (a part of the foaf ontology), and two mappings between them. Graphically, the mappings are described by the arrows that go between the mapped schema elements.

Mappings are often defined as views in conventional data integration systems, often in the form of GAV (global-as-view), LAV (local-as-view), or, more generally, GLAV (global-and-local-as-view) assertions [4]. We take the LAV approach, i.e., we define each relational table in the source as a view over the RDF ontologies. We call such views as RDF Views. For formal discussion, RDF views are expressed in a Datalog-like notation.

The lower part of Fig. 2 illustrates how to represent the semantic mappings as RDF views. A typical RDF view consists of two parts. The left part is called the view head, and is often a relational predicate. The right part is called the view body, and is often a set of RDF triples. In general, the body can be viewed as a RDF query over the target RDF ontology, and it defines the semantics of the relational predicate from the perspective of the RDF ontology.

³The Friend of a Friend (FOAF) project is about creating a Semantic Web of machine-readable homepages describing people, the links between them and the things they create and do: <http://www.foaf-project.org/>

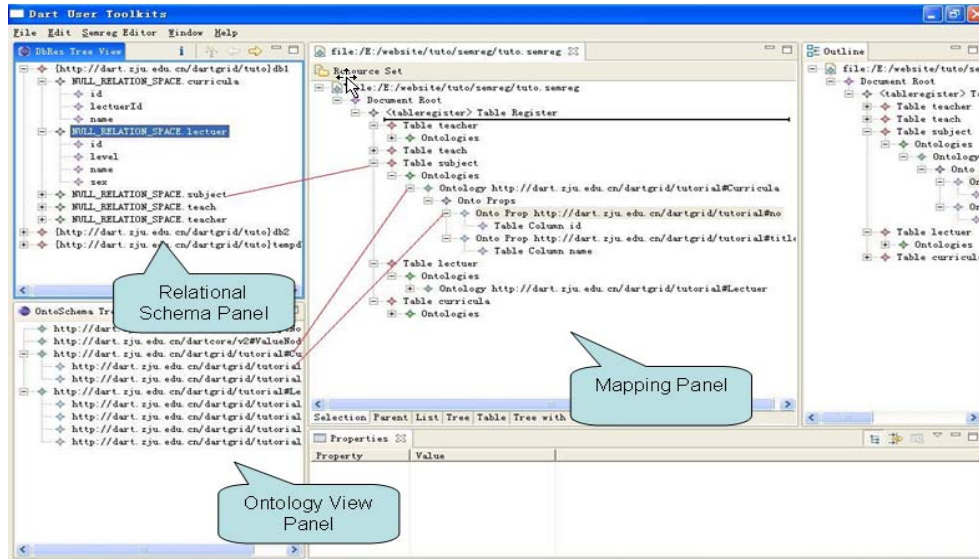


Figure 3. Visual Semantic Mapping Tool.

The semantics of RDF view would be more clear if we construct a *Target RDF Instance G* based on the semantic mapping specified by RDF views. For example, given a relational tuple as below, applying the RDF view V1 in Fig. 2 on this tuple will yield a set of RDF triples. We illustrate these RDF triples using N3 notation.

Relational Tuple:

```
w3c:emp("DanBrickley", "danbri@w3.org",
        "SWAD", "http://swad.org", "EU");
```

Yielded RDF triples by Applying V1:

```
_:bn1 rdf:type foaf:Person;
foaf:name "Dan Brickley";
foaf:mbox "danbri@w3.org";
foaf:currentProject _:bn2.
_:bn2 rdf:type foaf:Project;
foaf:name "SWAD";
foaf:homepage "http://swad.org";
foaf:fundedBy _:bn3.
_:bn3 rdf:type foaf:Organization;
foaf:name "EU".
```

The key notion is the newly generated blank node ID in the RDF triples. As can be seen, corresponding to each existential variable $?y \in \bar{Y}$ in the view, we generate a new blank node ID. For examples, $_:bn1, _:bn2$ are both newly generated blank node IDs corresponding to the variables $?y1, ?y2$ in V1. This treatment of existential variable is in

accordance with the RDF semantics, since blank nodes can be viewed as existential variables ⁴.

We formally define the RDF view as below.

Definition 1. RDF View Let Var be a set of variable names. A typical RDF View is like the form :

$$R(\bar{X}) : -G(\bar{X}, \bar{Y}), \text{ where :}$$

1. $R(\bar{X})$ is called the head of the view, and R is a relational predicate ;
2. $G(\bar{X}, \bar{Y})$ is called the body of the view, and G is a RDF graph with some nodes replaced by variables in Var .
3. The $\bar{X}, \bar{Y}$ contain either variables or constants. The variables in $\bar{X}$ are called distinguished variables, and the variables in $\bar{Y}$ are called existential variables. We often denote individual existential variables by $"?y1, ?y2, \dots"$.

For formal discussion on RDF view, please refer to another paper of ours [5].

3.2 Visual Mapping Tool

To speed up the process of defining RDF views, a *Visual Semantic Mapping* tool is developed. As Fig.3 displays, there are three major panels of the mapping tool.

⁴W3C Document RDF Semantics :<http://www.w3.org/TR/rdf-ml/>

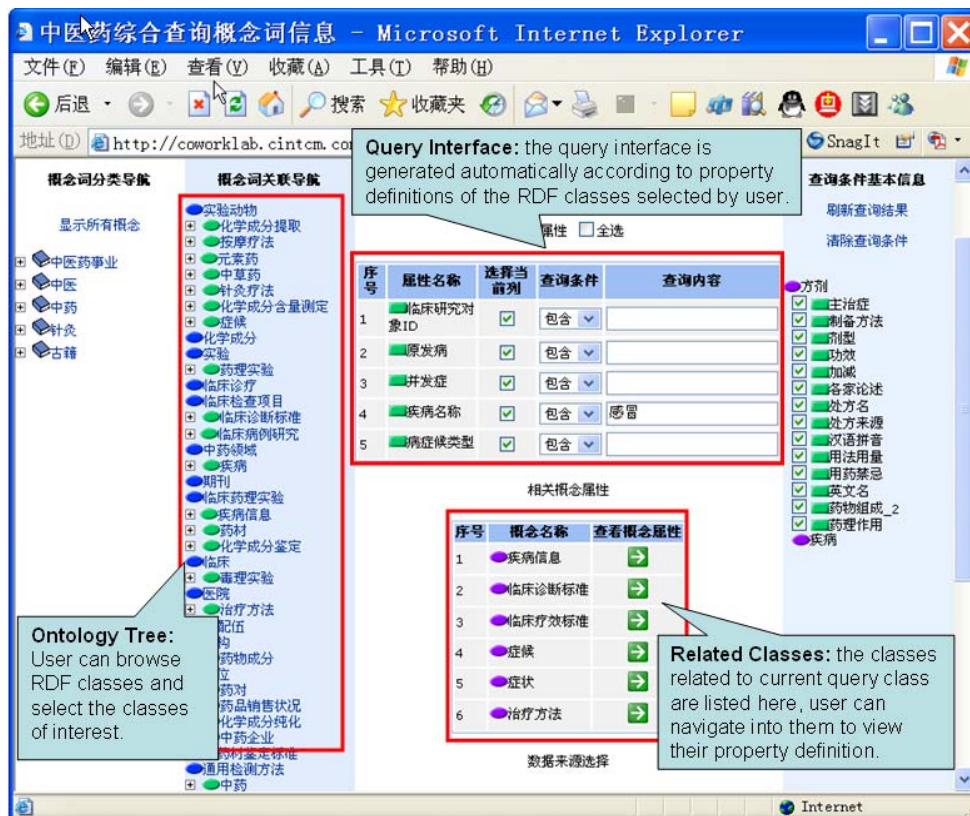


Figure 4. Semantic Query Tool.



Figure 5. Semantic Search and Navigation Tool.

1. Relational Schema Panel. This panel is used to display the relational table definitions.
2. Ontology Panel. This panel is used to display shared ontology definitions.
3. Mapping Panel. User drags table or column names from the relational schema panel, and drags class or property names from the ontology panel, and establishes the mappings from relational schema to shared ontology in the mapping panel.

A set of RDF views will be finally generated and submitted to the semantic registration service for future querying.

4 Semantic Query and Search

To query the system, programmer could manually define RDF queries, encode them to his program, and invoke the semantic query service to evaluate the queries. For normal user, DartGrid also offers a query tool and a search tool.

4.1 Semantic Query Tool

Fig. 4 displays a snapshot of the visual query tool. This tool help user to define complex RDF queries.

1. The tool retrieves ontology definitions from the ontology service, and display it as a ontology tree.
2. User browses the ontology and select the RDF class of interested.
3. The tool automatically generate the query interface based on the RDF class definition.
4. User selects the properties and input some constraints.
5. User navigates into related RDF class and selects more properties and input more constraints.
6. A RDF query is automatically generated and submitted to the semantic query service.

4.2 Semantic Search and Navigation Tool

DartGrid also provide user with a google-like search interface as Fig. 5 displays. User could search databases and semantically navigate the datas from one database to another database.

When a keyword is submitted, all of the relevant database entries are retrieved and displayed as semantically enriched format, i.e., RDF format. For each entry, the RDF classes that the data entry belongs to are listed below the data. More importantly, all of the names of those data entries relating to the current data entry are also listed as hyper

links, so that user could navigate into them to view all of the related information. Because the relationship is established at a semantic level, we call those links as semantic link and the navigation as semantic navigation.

5 Conclusions

DartGrid is a data integration toolkit using technologies from semantic web and grid. It is built upon Globus toolkit and offers basic grid services for building data-intensive grid application. It takes an ontology-based approach from semantic web to provide a semantic heterogeneity solution capable of deployment in grid settings. A set of semantic tools are developed for register, query, search and manage database resources. This toolkit is downloadable from <http://ccnt.zju.edu.cn/projects/dartgrid>.

6 Acknowledgement

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